Rule Based Fraud Detection

The Problem

Input

- Structured Dataset
  - Onboarding data (name, contact info)
  - Event data (login, device, registered devices, amount, time)
  - Transaction data (sender, recipient)

Output

- Ranked Fraudulent Events

What We Would like

Low false negative (high cost of human checking fraud)

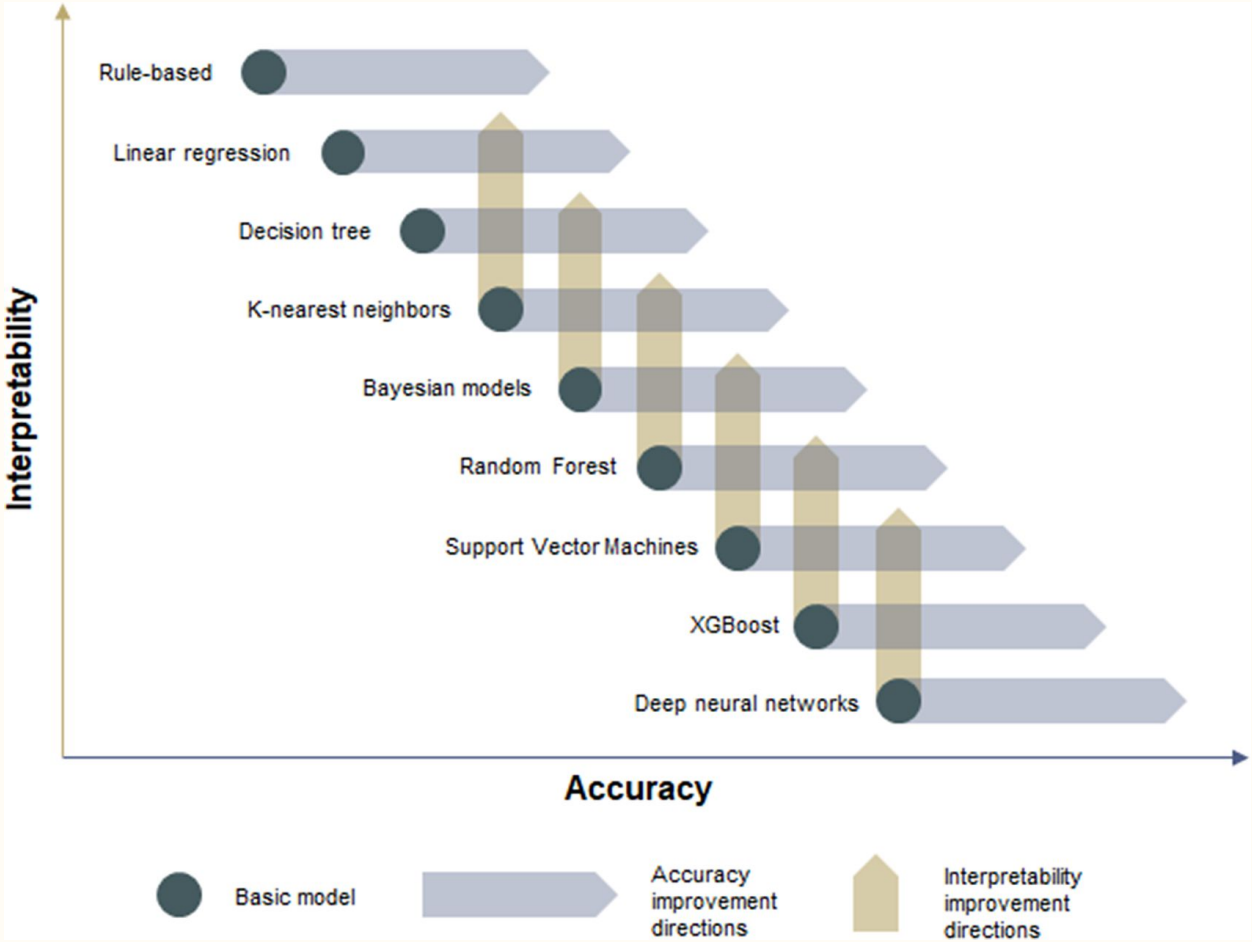
Interpretable model (to keep the clients and regulators happy)

Dataset Challenge - Imbalance Learning

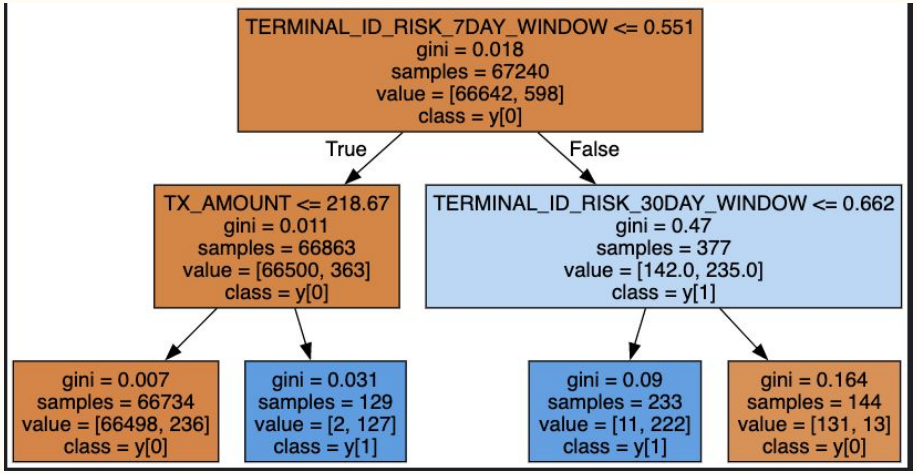
1) Financial Fraud Detection Dataset
   a) 6362620 transactions loaded, containing 8213 fraudulent transactions (0.12% negative labels)

2) Synthetic Dataset
   a) 919767 transactions loaded, containing 8195 fraudulent transactions (0.89% negative labels)

Model Challenge - Interpretability



Model Results - Synthetic Data



5 rows 5 rows × 3 columns

	AUC ROC	Average precision
Logistic regression	0.870	0.605
Decision tree with depth of two	0.763	0.496
Decision tree - unlimited depth	0.788	0.309
Random forest	0.867	0.658
XGBoost	0.875	0.605